2.DATA SPLITTING

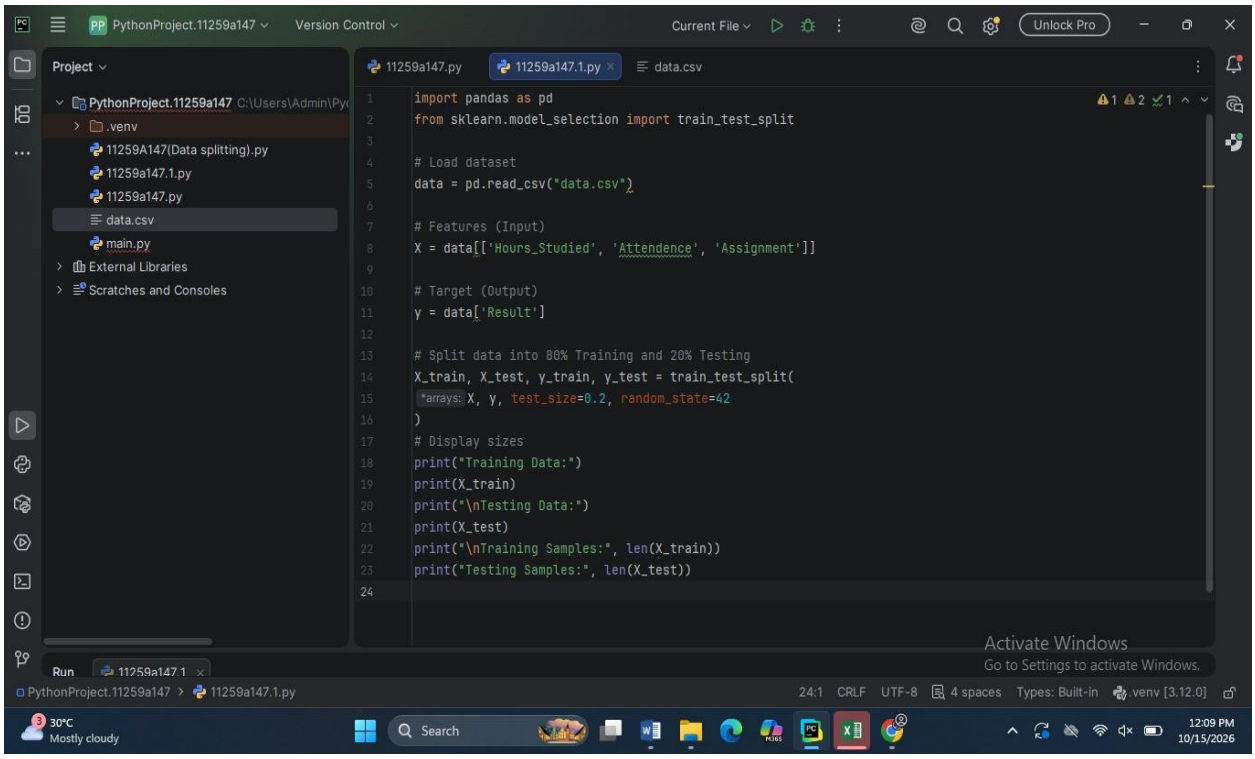
Aim

To write a program to split the given dataset into training and testing sets, so that a model can be trained on one part of the data and evaluated on unseen data.

Algorithm

1. Start the program.
2. Import the required libraries (pandas and train_test_split from sklearn.model_selection).
3. Load the dataset from the input file.
4. Separate the data into input features (X) and the target variable (y).
5. Choose the split ratio, for example 80% for training and 20% for testing.
6. Split the data into training and testing sets using train_test_split, with a fixed random_state so the split can be repeated.
7. Display the number of records in the training set and the testing set.
8. Display the training and testing data.
9. Stop the program.

Program:



```
import pandas as pd
from sklearn.model_selection import train_test_split

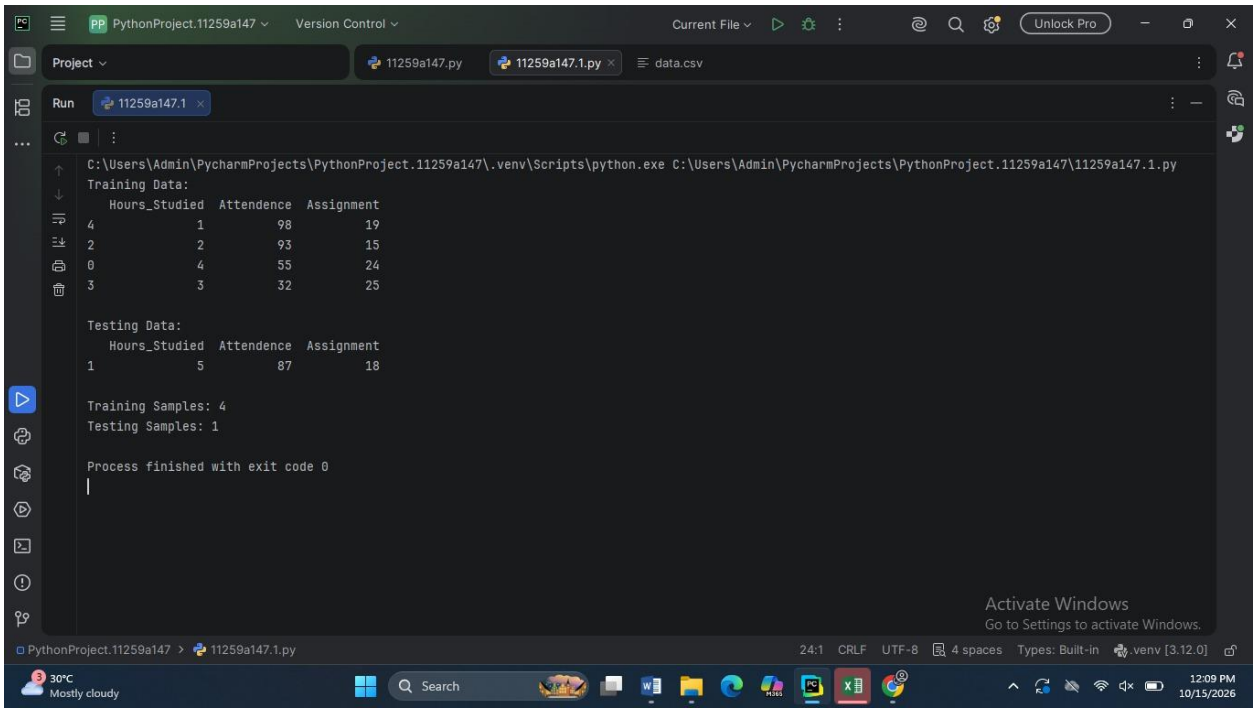
# Load dataset
data = pd.read_csv("data.csv")

# Features (Input)
X = data[['Hours_Studied', 'Attendence', 'Assignment']]

# Target (Output)
y = data['Result']

# Split data into 80% Training and 20% Testing
X_train, X_test, y_train, y_test = train_test_split(
    *arrays: X, y, test_size=0.2, random_state=42
)
# Display sizes
print("Training Data:")
print(X_train)
print("\nTesting Data:")
print(X_test)
print("\nTraining Samples:", len(X_train))
print("Testing Samples:", len(X_test))
```

Result:



```
C:\Users\Admin\PycharmProjects\PythonProject.11259a147\.venv\Scripts\python.exe C:\Users\Admin\PycharmProjects\PythonProject.11259a147\11259a147.1.py
Training Data:
   Hours_Studied  Attendence  Assignment
4              1          98          19
2              2          93          15
0              4          55          24
3              3          32          25

Testing Data:
   Hours_Studied  Attendence  Assignment
1              5          87          18

Training Samples: 4
Testing Samples: 1

Process finished with exit code 0
```